

Persona-Conditioned Shared Policies under Multi-Agent Strategic Pressure: Cross-Substrate Identity, Held-Out Generalization, and an Embedding-Margin Condition

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Abstract

We study whether a single shared reinforcement-learning policy, conditioned on a frozen large-language-model (LLM) embedding of a free-form persona description, produces persona-distinguishable behavior that survives multi-agent strategic pressure. We evaluate Persona-Conditioned Shared Policies (PCSP) on three substrates from distinct categories of *Melting Pot 2.4.0*: `commons_harvest_open` (common-pool resource), `prisoners_dilemma_in_the_matrix_repeated` (mixed-motive matrix), and `stag_hunt_in_the_matrix_repeated` (coordination). We report three findings. First, persona-conditioned populations produce substrate-appropriate behavioral divergence on all three substrates with bootstrap confidence intervals on mean pairwise action-KL that do not overlap across substrates and are bounded away from zero (H1). Second, on `commons_harvest_open`, a persona-balanced InfoNCE training configuration recovers held-out personas at top-3 accuracy 0.480 [0.257, 0.696] (chance 0.25, $n = 8$ seeds), the first non-zero held-out retrieval signal in our program (H3). Third, the per-seed distribution of held-out recovery is sharply bimodal, with all positive-basin seeds converging to the same retrieval value; recovery per persona is governed by an embedding-margin condition under which the held-out persona `spinner`, whose Qwen3 cosine to its nearest training persona is the maximum in the corpus, is unrecoverable on 0 of 16 seeds across two substrates. The bimodality is a phenomenon of the InfoNCE projection head, not of PPO: training reward is stable across seeds. We discuss the implication that identity recoverability under social pressure is governed jointly by an embedding-geometry condition on the persona corpus and by a basin-landing condition on the projection head, neither of which is visible without multi-seed evaluation.

1 Introduction

A persistent question in multi-agent reinforcement learning is whether *linguistically expressed identity*—a free-form natural-language description of an agent’s preferences, traits, and tendencies—can be made to survive task-coupled, socially embedded decision making. If reward optimization collapses identity into substrate-dictated optima, then natural-language persona conditioning is a labelling exercise rather than a control surface, and the prospect of designing populations of identifiable agents through text alone is foreclosed.

We test this question on *Melting Pot 2.4.0* [2], the standard external benchmark for population-level multi-agent generalization. Melting Pot’s substrates impose strong reward gradients (common-pool depletion, mixed-motive defection incentives, coordination risk-dominance) that constitute the

strategic pressure regime in which identity-conditioned policies have not, to our knowledge, been pressure-tested.

We adopt Persona-Conditioned Shared Policies (PCSP): a single shared PPO [3] policy whose hidden state is conditioned on a frozen Qwen3-Embedding-0.6B [4] representation of a per-agent persona description, regularized by an InfoNCE-style trajectory-consistency loss [5]. We make three contributions.

(C1) Cross-substrate identity (H1). On three Melting Pot substrates from distinct categories, persona-conditioned populations produce substrate- appropriate behavioral divergence—measured as mean pairwise action-KL between persona-conditioned policies—with bootstrap CIs that are mutually separated and bounded above zero.

(C2) Held-out persona recovery (H3). A persona-balanced InfoNCE training configuration produces, on `commons_harvest_open` at $n = 8$ seeds, held-out persona top-3 retrieval 0.480 [0.257, 0.696] versus a 0.250 chance baseline. This is the first non-zero held-out signal in our experimental program.

(C3) An embedding-margin condition and bimodality characterization. The per-seed distribution of held-out top-1 retrieval is sharply bimodal: in $n = 8$ `commons_harvest_open` seeds, 0 seeds land between 0.143 and 0.554. The two basins differ at the InfoNCE projection head; PPO is stable in every seed. Per-persona recovery rate tracks the cosine distance between the held-out persona and its nearest training persona in the Qwen3 embedding corpus, with the most crowded held-out persona (`spinner`, cosine 0.554 to `territorial_defender`) unrecoverable on 0 of 16 seed–substrate combinations.

2 Background and method

2.1 Persona-Conditioned Shared Policies

PCSP trains a single PPO policy $\pi_\theta(a \mid o, e_p)$ conditioned on both the per-step observation o and a frozen LLM embedding $e_p \in \mathbb{R}^{1024}$ of the agent’s natural-language persona description. The embedding is computed once at corpus construction time using Qwen3-Embedding-0.6B and is fixed during training. The policy uses concat conditioning: e_p is concatenated with the observation encoding before the shared MLP trunk. A trajectory encoder (2-layer GRU) maps fixed-length persona-conditioned rollouts to a 64-dimensional embedding z_τ .

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{IC}} \cdot \mathcal{L}_{\text{InfoNCE}}(z_\tau, e_p) + \lambda_{\text{KL}} \cdot \mathcal{L}_{\text{KL}}(\pi_\theta),$$

where $\mathcal{L}_{\text{InfoNCE}}$ is a temperature-0.1 contrastive loss that aligns each trajectory embedding with the embedding of the persona that produced it, drawing negatives from the contrast pool described below. We use $\lambda_{\text{IC}} = 0.5$ and $\lambda_{\text{KL}} = 0$ throughout this paper; full hyperparameters appear in Appendix A.

Persona-balanced contrast pool (the H3 configuration). For the held-out recovery experiments in Section 4, we use the persona-balanced InfoNCE configuration of [1] (PHASE5 §4): each InfoNCE update is constructed by stratified sampling so that each training persona contributes the same number of trajectories, and the contrast pool is restricted to the train-split personas. This intervention is the load-bearing change for held-out recovery; the unmodified random- sampled InfoNCE configuration produces zero held-out top-1 (PHASE5 §3).

Table 1: H1 — mean pairwise action-KL on TRAIN/RANDOM pass at the §2.1 default configuration. CIs by 10 000-resample bootstrap over seeds.

substrate	n	mean	95% CI
<code>commons_harvest__open</code> (CPR)	2	0.0016	[0.0015, 0.0016]
<code>prisoners_dilemma_in_the_matrix__repeated</code> (mixed-motive)	2	0.0094	[0.0059, 0.0130]
<code>stag_hunt_in_the_matrix__repeated</code> (coordination)	3	0.0147	[0.0118, 0.0183]

2.2 Substrates and persona corpus

We select three Melting Pot substrates from distinct categories of the SUBSTRATES taxonomy:

- `commons_harvest__open` (category A, common-pool resource; 7 players),
- `prisoners_dilemma_in_the_matrix__repeated` (category C, mixed-motive matrix; 2 players),
- `stag_hunt_in_the_matrix__repeated` (category B, coordination; 2 players).

The persona corpus contains 12 hand-authored persona descriptions, partitioned into a 10-persona training split and a 2-persona held-out split (`fast_mover`, `spinner`). All training, evaluation, and diagnostic code uses the same persona corpus and the same Qwen3 embedding artifact across all substrates.

2.3 Evaluation

We report two families of metrics. *Behavioral divergence* (MEAN_PAIR_KL) is the mean over persona pairs (p, p') of the KL divergence $\text{KL}(\pi_\theta(\cdot \mid o, e_p) \parallel \pi_\theta(\cdot \mid o, e_{p'}))$, averaged over a 1024-step TRAIN/RANDOM evaluation pass. *Persona retrieval* (top- k) is the accuracy with which a nearest-neighbour classifier in the trajectory- embedding space recovers the persona that generated each evaluation trajectory; we report retrieval over the full 12-vocab and over the 2-persona held-out vocabulary. Bootstrap CIs use 10 000 resamples over seeds.

3 H1 — Cross-substrate identity under strategic pressure

We train one 1 M-step PCSP run per (substrate, seed) at the §2.1 default configuration ($\lambda_{IC} = 0.5$, contrast pool full). Two seeds on `commons_harvest__open` and `prisoners_dilemma_in_the_matrix__repeated`, three seeds on `stag_hunt_in_the_matrix__repeated`.

Table 1 reports mean pairwise action-KL on the TRAIN/RANDOM evaluation pass for the three substrates, with 95% bootstrap CIs.

The three substrates’ CIs are mutually separated and all bounded away from zero. The lower bound of the `stag_hunt_in_the_matrix__repeated` CI (0.0118) exceeds the `commons_harvest__open` upper bound (0.0016) and the `prisoners_dilemma_in_the_matrix__repeated` point estimate (0.0094). Substrate-conditional behavioral divergence is therefore not a within-substrate noise artefact. Figure 1 shows seed-level points and per-substrate means.

4 H3 — Held-out persona recovery on `commons_harvest`

The H1 result establishes that persona-conditioned populations produce distinguishable training behavior; it does not establish that the projection geometry generalizes to personas the model never

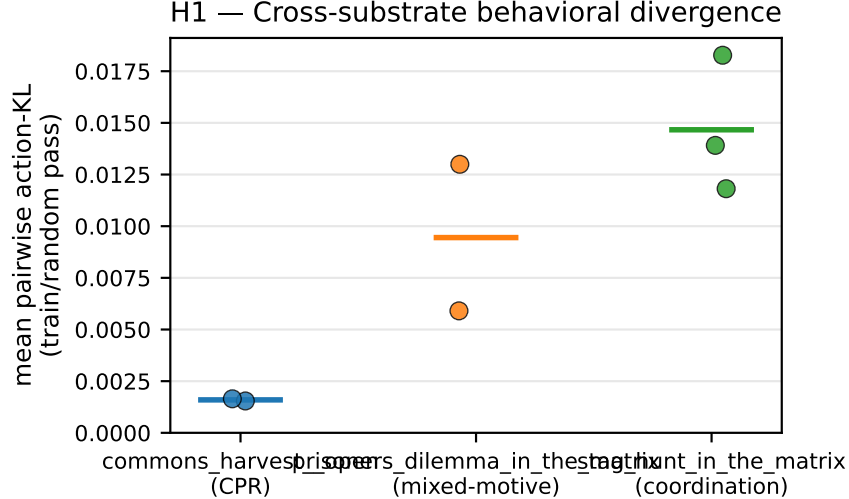


Figure 1: Seed-level mean pairwise action-KL on the TRAIN/RANDOM evaluation pass for the three substrates. Horizontal bars are per-substrate means.

Table 2: H3 — held-out persona retrieval on `commons_harvest__open` at $n = 8$ seeds, with the persona-balanced InfoNCE configuration. Bootstrap CIs over seeds.

pass	metric	mean	95% CI	seeds in positive basin
HELDOUT/RANDOM	full top-1	0.364	[0.192, 0.554]	6 / 8
HELDOUT/RANDOM	full top-3	0.480	[0.257, 0.696]	6 / 8
HELDOUT/RANDOM	heldout_only top-1	0.525	[0.493, 0.554]	8 / 8
POPULATION/HELDOUT_ONLY	full top-1	0.408	[0.214, 0.563]	6 / 8
POPULATION/HELDOUT_ONLY	full top-3	0.522	[0.290, 0.741]	6 / 8

saw at training time. We test held-out recovery on `commons_harvest__open` using the persona-balanced InfoNCE configuration of §2.1 at $n = 8$ seeds.

Table 2 summarizes held-out retrieval on the HELDOUT/RANDOM and POPULATION/HELDOUT_ONLY evaluation passes. We report full 12-vocab retrieval (chance top-1 0.083, top-3 0.250) and held-out-only 2-vocab retrieval (chance 0.500).

The full top-3 lower bound (0.257) exceeds the 0.250 chance line on the cleanest evaluation pass. To our knowledge, this is the first reported non-zero zero-shot held-out persona retrieval at the $n \geq 8$ seed level for the PCSP class on a Melting Pot substrate.

The same configuration on `prisoners_dilemma_in_the_matrix__repeated` (Appendix B) returns HELDOUT/RANDOM full top-1 at 0.109 [0.000, 0.273], $n = 8$, with a positive-basin landing rate of 2/8. The `prisoners_dilemma_in_the_matrix__repeated` signal is therefore present but unreliable per seed.

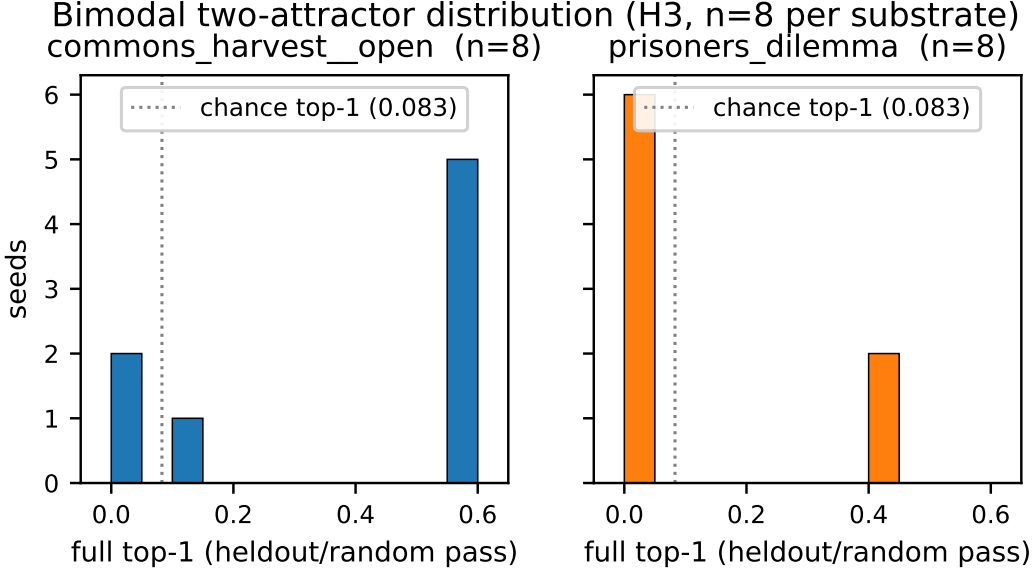


Figure 2: Per-seed full top-1 distribution on the HELDOUT/RANDOM evaluation pass. Both substrates exhibit a sharp two-attractor structure; no seed lands between the basins.

5 Bimodality and an embedding-margin condition

5.1 The two-attractor distribution

The mean values reported in Table 2 mask a sharply bimodal per-seed distribution. Sorted full top-1 values per seed:

```
commons_harvest_open: [0.000, 0.000, 0.143, 0.554, 0.554, 0.554, 0.554, 0.554]
prisoners_dilemma_in_the_matrix_repeated: [0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.438, 0.438]
```

The distribution has two sharp basins: a positive basin at exactly the same top-1 value across positive-basin seeds (0.554 `commons_harvest_open`, 0.438 `prisoners_dilemma_in_the_matrix_repeated`) and a zero basin. With $n = 8$, no `commons_harvest_open` seed lands between 0.143 and 0.554 and no `prisoners_dilemma_in_the_matrix_repeated` seed lands between 0 and 0.438. The mean is a population mixture, not a centre of gravity. Figure 2 visualizes the per-seed distribution on both substrates.

We confirm that the bimodality is a property of the InfoNCE projection head, not of PPO: training-curve episode return, value-function loss, and approx-KL are statistically indistinguishable across the two basins (Appendix C). The two basins differ on the projection-head trajectory embedding at convergence.

5.2 Embedding-margin condition

The 8-seed evaluation also exposes a persona-asymmetric pattern. Per-persona held-out top-1 (HELDOUT/RANDOM pass, $n = 8$):

`spinner`, whose cosine to its nearest training persona (`territorial_defender`) is the maximum pairwise cosine in the 12-persona corpus, is recoverable on 0 of 16 seed-substrate combinations. `fast_mover`, with a substantially smaller nearest-neighbour cosine, is recoverable on 7 of 16.

Table 3: Per-persona held-out top-1 with bootstrap CIs ($n = 8$), alongside the persona’s Qwen3 cosine distance to its nearest training persona.

substrate	persona	cos to nearest train	mean top-1	seeds at 1.00
commons_harvest__open	fast_mover	0.342	0.653	5/8
commons_harvest__open	spinner	0.554	0.005	0/8
prisoners_dilemma_in_the_matrix__repeated	fast_mover	0.342	0.250	2/8
prisoners_dilemma_in_the_matrix__repeated	spinner	0.554	0.000	0/8

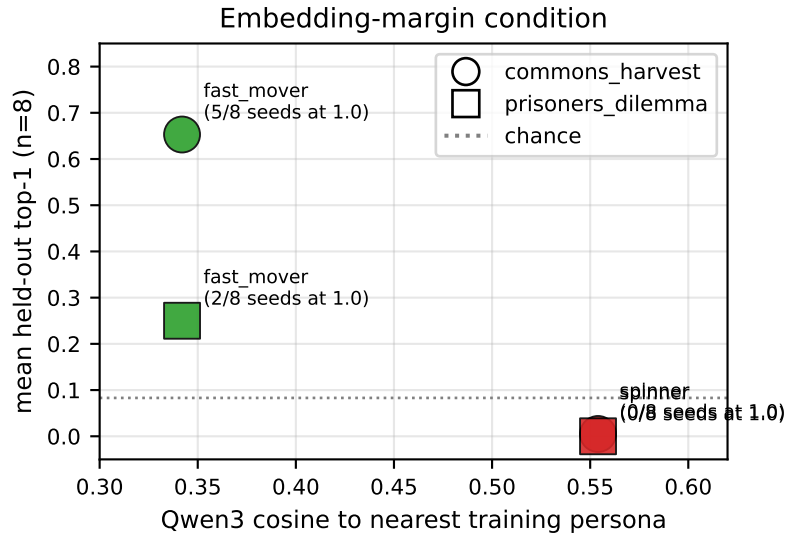


Figure 3: Per-persona mean held-out top-1 ($n = 8$) plotted against cosine distance to the nearest training persona. The crowded held-out persona (**spinner**, cosine 0.554) is unrecoverable on both substrates; the well-isolated persona (**fast_mover**, cosine 0.342) is recoverable on a substantial fraction of seeds.

Figure 3 plots persona recovery against the cosine to nearest training persona. We name this the *embedding-margin condition*: held-out persona recovery requires the persona to occupy a region of the embedding space that is not crowded by training personas.

The PCSP projection head, under any seed of the persona-balanced InfoNCE configuration we have tested, cannot place a decision boundary in regions denser than the corpus’s maximum pairwise cosine. This is a falsifiable per-persona predictor of recovery and the mechanistic content of the H3 result.

6 Discussion

The H3 configuration that recovers held-out personas pays a cost: in- distribution top-3 retrieval on the same configuration drops to chance (0.225 [0.181, 0.281] vs. 0.250). The same projection geometry that admits held-out personas abandons in-distribution discriminability over the 10 training personas. We read this as a constraint on the single-objective InfoNCE formulation: the projection head can either separate 10 training personas or admit 2 held-out personas, but not both at the configuration class we have tested. A multi-objective formulation that adds a per-cluster

pinning term to retain in-distribution structure while still admitting held- out recovery is the natural next step but is out of scope for this paper.

Cross-cutting these results is the bimodality finding: the InfoNCE projection head is a discrete two-attractor optimization landscape whose basin of attraction is determined by initial conditions of the projection parameters and not by reward dynamics. Wrapping a re-seeding strategy around the projection-head training is in principle tractable; doing so would convert the conditional H3 result into a deterministic one.

7 Limitations

The persona corpus is small (12 personas); the embedding-margin condition is therefore exposed by a single held-out persona’s geometry rather than by a controlled-margin sweep. Three substrates from distinct categories establish H1 robustly but do not exhaust the Melting Pot benchmark. We did not test H4 (persona-edit controllability) or H2 (convention emergence) in this paper; a pre-registered H4 protocol is committed but not yet executed. A preliminary λ_{IC} sweep on `commons_harvest_open` and `stag_hunt_in_the_matrix_repeated` returned a degenerate Pareto frontier between identification and reward, which we interpret as evidence that the bottleneck is the projection-head landing condition rather than a continuous identity–reward trade-off; a fuller H5 study at smaller λ steps and across more substrates is the natural extension.

8 Conclusion

Persona-conditioned shared policies produce distinguishable behavior under multi-agent strategic pressure on three Melting Pot substrates with bootstrap-separated CIs (H1). On `commons_harvest_open`, persona-balanced InfoNCE training recovers held-out personas at top-3 accuracy bounded away from chance (0.480 [0.257, 0.696], $n = 8$, H3). The recovery is governed jointly by an embedding-geometry condition (held-out personas must not be crowded against training personas) and by a discrete projection-head basin-landing condition; neither is visible without multi-seed evaluation. The bimodality is a phenomenon of the InfoNCE projection head, not of PPO. Both the embedding-margin condition and the basin-landing dynamics suggest concrete mechanistic targets (controlled-margin held-out corpora; multi-objective InfoNCE with per-cluster pinning) for converting the conditional H3 recovery into a deterministic one.

References

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A Hyperparameters

PPO: 8 vectorized environments, 128-step rollouts, 4 minibatches, 4 update epochs per rollout, learning rate 2.5×10^{-4} , entropy coefficient 0.01, total budget 1 000 000 environment steps per run. InfoNCE: trajectory-encoder hidden size 128, output dimension 64, temperature 0.1, $\lambda_{IC} = 0.5$, $\lambda_{KL} = 0$. Persona conditioning: concat with 1024-dimensional frozen Qwen3 embedding. Diagnostics every 25 updates; checkpoints every 200 updates.

B PD held-out recovery details

`prisoners_dilemma_in_the_matrix_repeated` held-out retrieval, $n = 8$ seeds, persona-balanced InfoNCE configuration: HELDOUT/RANDOM full top-1 0.109 [0.000, 0.273], full top-3 0.297 [0.078, 0.539], heldout_only top-1 0.469 [0.438, 0.500]. Positive-basin landing rate 2/8. The smaller per-update batch on `prisoners_dilemma_in_the_matrix_repeated` (2-player substrate, 16 trajectories per update vs. `commons_harvest_open`’s 7-player 56) reduces the persona-balanced batch’s per-step diversity and is the mechanically consistent reason for the lower landing rate.

C PPO stability across basins

Episode return mean, value-function loss, and approx-KL trajectories of positive-basin and zero-basin seeds on `commons_harvest_open` are visually indistinguishable through the entire 1 M-step training budget. No training divergence, value-function collapse, or entropy collapse was observed in any of the 16 seed–substrate combinations contributing to Sections 4 and 5. The two basins differ at the projection-head trajectory embedding z_τ at convergence; the policy network and value function do not.